

Conversion Log

Production Conversion Log

April 2025 · Girardot, Colombia

Submission reference	RTF0-310825
Reporting period	April 2025 (2025-04)
Plant	Girardot, Colombia
Product	DEV-P100 — refined pyrolysis oil
Off-taker	Crown Oil Limited (UK)
Regulator	UK DfT — LCF Delivery Unit
Density — EU-grade	0,778 kg/L
Density — PLUS-grade	0,865 kg/L
Production days observed	24 of 30 (6 non-production days)
Source	daily_production table (DFT backend, snapshot 2026-05-21T00:00:00Z)
Generated	2026-05-21T00:00:00Z

Signed for and on behalf of OisteBio GmbH

Paolo Ughetti

Managing Director

Date: _____

Executive Summary

Per-day kilogram-to-litre conversion log for **April 2025** production at the Girardot, Colombia pyrolysis facility, operated by **OisteBio GmbH**. Output product is **DEV-P100 — refined pyrolysis oil**, supplied to **Crown Oil Limited (UK)** under the UK DfT — LCF Delivery Unit reporting framework. Litres are derived from kilograms via the nominal product densities listed below, as materialised by the `litres_eu` and `litres_plus` GENERATED ALWAYS columns on the `daily_production` table (Alembic migration `0007_persist_production_litres.py`).

Monthly aggregate

Metric	Value
Feedstock routed to thermal conversion (input)	2 959 166,000 kg
Total EU-grade production	869 994,804 kg / 1 118 245,249 L
Total PLUS-grade production	1 197 791,487 kg / 1 384 730,039 L
Liquid-product yield (EU + PLUS) over input	69,88 %
Grand total volume (EU + PLUS)	2 502 975,288 L

Co-products tracked separately on the same daily ledger (not converted to litres because they are solid / gaseous): carbon black **430 595,258 kg**, steel scrap **240 631,881 kg**, water condensate **43 518,534 kg**, syngas **121 595,899 kg**, recorded losses **55 038,137 kg**. Mass-balance closure across all streams is reported in Annex A.

Daily Conversion — April 2025

One row per production day from the `daily_production` table. Days with no production record (weekends, public holidays, maintenance) are listed in the gap line below. Litres computed at constant density: **EU 0,778 kg/L, PLUS 0,865 kg/L**. Liquid yield = (EU kg + PLUS kg) / input kg.

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-04-01	131 181,000	38 567,214	53 098,571	0,778	0,865	49 572,254	61 385,631	69,88 %	—
2025-04-02	120 244,000	35 351,736	48 671,565	0,778	0,865	45 439,249	56 267,705	69,88 %	—
2025-04-03	119 435,000	35 113,890	48 344,103	0,778	0,865	45 133,535	55 889,136	69,88 %	—
2025-04-04	116 897,000	34 367,718	47 316,788	0,778	0,865	44 174,445	54 701,489	69,88 %	—
Total	2 959 166,000	869 994,804	1 197 791,487	—	—	1 118 245,249	1 384 730,039	69,88 %	

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-04-05	114 235,000	33 585,090	46 239,282	0,778	0,865	43 168,496	53 455,817	69,88 %	—
2025-04-07	118 859,000	34 944,546	48 110,954	0,778	0,865	44 915,869	55 619,600	69,88 %	—
2025-04-08	112 995,000	33 220,530	45 737,363	0,778	0,865	42 699,910	52 875,564	69,88 %	—
2025-04-09	104 943,000	30 853,242	42 478,128	0,778	0,865	39 657,123	49 107,662	69,88 %	—
2025-04-10	110 610,000	32 519,340	44 771,978	0,778	0,865	41 798,638	51 759,512	69,88 %	—
2025-04-11	124 396,000	36 572,424	50 352,184	0,778	0,865	47 008,257	58 210,617	69,88 %	—
2025-04-12	128 564,000	37 797,816	52 039,279	0,778	0,865	48 583,311	60 161,016	69,88 %	—
2025-04-14	138 193,000	40 628,742	55 936,841	0,778	0,865	52 222,033	64 666,868	69,88 %	—
2025-04-15	130 864,000	38 474,016	52 970,257	0,778	0,865	49 452,463	61 237,291	69,88 %	—
2025-04-16	124 749,000	36 676,206	50 495,069	0,778	0,865	47 141,653	58 375,802	69,88 %	—
2025-04-19	113 832,000	33 466,608	46 076,158	0,778	0,865	43 016,206	53 267,235	69,88 %	—
2025-04-21	127 842,000	37 585,548	51 747,032	0,778	0,865	48 310,473	59 823,158	69,88 %	—
2025-04-22	133 280,000	39 184,320	53 948,190	0,778	0,865	50 365,450	62 367,850	69,88 %	—
2025-04-23	118 618,000	34 873,692	48 013,403	0,778	0,865	44 824,797	55 506,824	69,88 %	—
2025-04-24	127 833,000	37 582,902	51 743,390	0,778	0,865	48 307,072	59 818,948	69,88 %	—
2025-04-25	127 173,000	37 388,862	51 476,240	0,778	0,865	48 057,663	59 510,104	69,88 %	—
2025-04-26	121 488,000	35 717,472	49 175,103	0,778	0,865	45 909,347	56 849,830	69,88 %	—
2025-04-28	139 919,000	41 136,186	56 635,480	0,778	0,865	52 874,275	65 474,543	69,88 %	—
2025-04-29	120 276,000	35 361,144	48 684,517	0,778	0,865	45 451,342	56 282,679	69,88 %	—
2025-04-30	132 740,000	39 025,560	53 729,612	0,778	0,865	50 161,388	62 115,158	69,88 %	—
Total	2 959 166,000	869 994,804	1 197 791,487	—	—	1 118 245,249	1 384 730,039	69,88 %	

Non-production days (6): 2025-04-06, 2025-04-13, 2025-04-17, 2025-04-18, 2025-04-20, 2025-04-27. These dates have no row in daily_production and are reported as N/A by omission. Weekends and public holidays in Colombia during April 2025 account for all of them.

Methodology & References

Volumetric conversion applies the canonical relation `litres = kg / density` at the calibrated product density measured at 15 °C for the reporting month. Densities are resolved per production month from the platform `product_densities` table (one row per (product, effective_from) pair), so historical conversions remain reproducible if downstream batches recalibrate the product specification. The densities applied to April 2025 are:

- **EU-grade (DEV-P100-EU)** — $\rho_{\text{EU}} = 0,778 \text{ kg/L}$
- **PLUS-grade (DEV-P100-PLUS)** — $\rho_{\text{PLUS}} = 0,865 \text{ kg/L}$

Both density values are ISCC-recognised product-specification figures for the OisteBio DEV-P100 — refined pyrolysis oil family. They are seeded and maintained in the platform `product_densities` table — initially loaded by Alembic migration `0005_seed_product_densities.py` and extended with per-month effective rows by `0014_monthly_densities.py`. Once a reporting period closes, the row applicable to that month becomes immutable for audit purposes; new effective rows are appended forward, never back-dated.

Per-row litre values shown in the daily table are not stored independently — they are derived deterministically inside PostgreSQL as **GENERATED ALWAYS** columns on the `daily_production` table:

```
litres_eu    NUMERIC GENERATED ALWAYS AS (eu_prod_kg / 0.778) STORED,
litres_plus  NUMERIC GENERATED ALWAYS AS (plus_prod_kg / 0.865) STORED
```

These columns are introduced by Alembic migration `0007_persist_production_litres.py`. Once persisted, historical litre values become immutable for audit purposes even if product densities are revised in the future. Write access to `litres_eu` and `litres_plus` is prohibited at the schema level: any attempt to update them raises PostgreSQL error code 42601.

The **input kg** column reports the daily value of `daily_production.kg_to_production` — the mass of pre-conditioned ELT feedstock routed to the pyrolysis reactor that calendar day. The **yield %** column reports the fraction of input mass that exits the refining line as liquid product (EU + PLUS), with the remainder distributed across carbon black, steel scrap, water condensate, syngas, and recorded losses (totals reported in the Executive Summary above and reconciled in Annex A).

This Production Conversion Log accompanies Annex A (mass balance) and Annex D (closing-stock carry-over) in the RTFO-310825 submission bundle and is cryptographically anchored to the cover letter via the SHA-256 digest reported in the running footer (short prefix 000000000000). The authoritative digest is reproduced in `MANIFEST.sha256` at the bundle root.

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